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Lithium: An Old Drug for New Therapeutic Strategy for Alzheimer's Disease and Related Dementia

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Keywords

Lithium · Calcium dyshomeostasis · Neuronal nitric oxide · Clinical efficacy · Alzheimer's disease

Abstract

Background: Although Alzheimer's disease (AD) is the most common form of dementia, the effective treatment of AD is not available currently. Multiple trials of drugs, which were developed based on the amyloid hypothesis of AD, have not been highly successful to improve cognitive and other symptoms in AD patients, suggesting that it is necessary to explore additional and alternative approaches for the disease-modifying treatment of AD. The diverse lines of evidence have revealed that lithium reduces amyloid and tau pathology, attenuates neuronal loss, enhances synaptic plasticity, and improves cognitive function. Clinical studies have shown that lithium reduces the risk of AD and deters the progress of mild cognitive impairment and early AD. ***Summary:*** Our recent study has revealed that lithium stabilizes disruptive calcium homeostasis, and subsequently, attenuates the downstream neuropathogenic processes of AD. Through these therapeutic actions, lithium produces therapeutic effects on AD with potential to modify the disease process. This review critically analyzed the preclinical and clinical studies for the therapeutic effects of lithium on AD. We suggest that disruptive calcium homeostasis is likely to be the early neuropathological mechanism of AD, and the stabilization of disruptive calcium homeostasis by lithium would be associated with its therapeutic effects on neuropathology and cognitive deficits in AD. ***Key Messages:*** Lithium is likely to be efficacious for AD as a disease-modifying drug by acting on multiple neuropathological targets including disruptive calcium homeostasis.

Introduction

Alzheimer's disease (AD) is the most common form of dementia and currently the 6th leading cause of death in the US [1]. The cause of AD is poorly understood, and the effective treatment and prevention of AD is not available currently. Multiple clinical trials of drugs, which were developed based on the amyloid theory of AD to remove toxic beta-amyloid (Aβ) plagues, have shown inconsistent results and mostly failed to improve cognitive function in AD patients [2–4]. It is generally

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agreed that it is necessary to explore additional and alternative approaches for the development of disease-modifying treatment of the late-onset AD and related dementia.

Lithium, a clinical drug used for the treatment of bipolar disorder and other mental illness for 5 decades, has been considered as a candidate for treating AD, based on diverse lines of evidence of its therapeutic potential [5–8]. Lithium is known as a multifaceted drug with a number of benefits suitable for the treatment of AD and dementia. Lithium reduces amyloid and tau pathology, attenuates neuronal cell death, enhances synaptic plasticity, and improves cognitive function in various AD animal models [9–11]. Furthermore, lithium has been shown to enhance cognitive function and synaptic plasticity in normal aging [12, 13]. In addition to these animal studies, the numbers of clinical studies have revealed that lithium reduces the risk of AD and deters the progress of mild cognitive impairment (MCI) and early AD [7, 8]. Here, we discussed the preclinical and clinical studies for the broad therapeutic effects of lithium on AD.

Lithium Improves Cognitive Function and Upregulates Synaptic Plasticity in Normal Aging

Preclinical studies suggest that lithium may improve cognitive function in normal aging. Our study revealed that 4 weeks lithium treatment enhances learning and memory in healthy aged rats [13]. We also reported that the same lithium treatment magnifies long-term potentiation (LTP) in the hippocampus, a synaptic activity likely underlying learning and memory [14]. These findings are consistent with other studies, which reported that 4-week lithium treatments increased LTP in the hippocampus [15, 16]. The same lithium treatment increased the density of dendritic branches in the dendritic trees of the dentate gyrus granule cells and CA1 pyramidal cells, where branches are highly distributed and receive major afferent input related to memory formation [17, 18].

Thus, these studies have demonstrated that lithium treatment upregulates functional as well as structural synaptic plasticity in the hippocampus. Since hippocampus synaptic plasticity is critical for learning and memory [19–21], the positive actions of lithium in functional and structural synaptic plasticity in the hippocampus likely to be associated with its enhancing action in cognitive function. Interestingly, a recent study reported that chronic lithium treatment alone or with enriched environment promoted maintenance of LPT and memory retention in a specially bred mouse strain with accelerated aging [12].

Lithium also upregulates the expression and production of neurotrophic and neuroprotective factors, which are involved in memory formation and LTP generation such as phospho-cyclic adenosine monophosphate (cAMP) response element-binding protein (p-CREB), brain-derived neurotrophic factor (BDNF), and B-cell lymphoma 2 (BCl-2) [22–25]. These findings together suggest that lithium treatment may improve learning and memory by upregulating hippocampal synaptic plasticity and neuroactive molecules involved in learning and memory formation such BDNF and CREB in normal aging animals [26, 27].

Lithium Reduces Neuropathology and Enhances Synaptic Plasticity and Cognitive Function in AD Animal Models

A large body of evidence suggests the therapeutic potential of lithium for AD [5–7]. Lithium has been shown to reduce major neuropathology, improve synaptic plasticity, and cognitive function in diverse transgenic AD models (Table 1). Most studies using APP transgenic AD mouse models have consistently reported that lithium reduces Aβ aggregation, senile plaques, phosphorylated tau (p-tau) formation, neurofibrillary tangles, and neuronal death and improves learning and working memory [28–35]. Studies using p-tau and p-tau/APP transgenic mouse models have also reported that lithium reduces Aβ aggregation, p-tau formation, and neurofibrillary tangles [9, 35–38]. These studies have suggested that the therapeutic effects of lithium are likely to be associated with its inhibition of glycogen synthase kinase (GSK) 3β [31–33, 35, 39]. Indeed, several studies using a GSK-3β overexpression mouse model have demonstrated that lithium acts against Aβ and p-tau formation, enhances hippocampal synaptic plasticity, and improves cognitive function which was associated with normalizing dysregulated GSK-3β activity [6, 40–42].

Molecular Mechanism of Action of Lithium for AD

The therapeutic mechanism of action of lithium for AD is highly complex because lithium has diverse neuroprotective and neurotrophic actions. A large body of evidence suggests that the therapeutic targets of lithium for AD may include GSK-3, inositol monophosphatase (IMPase) and inositol polyphosphatase (IPPase), cAMP,

Table 1. Therapeutic effects of lithium on AD neuropathology, synaptic plasticity, and memory

	GSK-3β	Ca^{2+} signal	nNOS	Aβ	Senile plaque	Tau-P	NFT	Cell loss	Hipp. synapsis	Memory
APP transgenic										
Phiel et al. [31] 2003	↓			↓						
Su et al. [33] 2004	↓			↓						
Rockenstein et al. [32] 2006	↓			↓						↑
Zhang et al. [35] 2011				↓	↓					↑
Trujillo-Estra et al. [34] 2013				↓	↓			↓		
Nunes et al. [30] 2015				↓	↓					
Liu et al. [29] 2020					↓	↓				↑
TAU transgenic										
Perez et al. [36] 2003	↓			↓		↓	↓			
Leroys et al. [28] 2010						↓	↓	↓		→
APP/TAU transgenic										
Caccamo et al. [9] 2007				→		↓				
Schaeffer et al. [47] 2017								→		
Wiseman et al. [38] 2022		↓	↓			↓			↓	
GSK-3β transgenic										
Hooper et al. [42] 2007	↓							↓	↑	
GSK-3β/TAU transgenic										
Engel et al. [40] 2008					↓	↓	→			
Hernandez et al. [41] 2013					↓	↓			↑	↑

and CREB [6, 11, 43, 44]. Among the various mechanisms, the inhibition of GSK-3β activity may play a key role in improving AD neuropathology, synaptic plasticity, and cognitive function as lithium is a well-established inhibitor that can block GSK-3β directly as well as indirectly [45, 46]. It is known that overactivation of GSK-3β is related with neuropathology of AD. Studies with AD mice with GSK-3β overexpression (Tet/GSK-3β mice) and APP/PS1 have shown that GSK-3β dysregulation accelerates p-tau and Aβ deposit, and lithium reduces the neuropathology and improve cognitive function [9, 32, 35, 37, 40, 41]. These molecular changes are associated with AD-like neurodegeneration, impaired hippocampal synaptic plasticity, and cognitive deficits [35, 47]. Lithium blocks excessive GSK-3β activity, increases the expression of BDNF, Bcl-2, and other neuroplastic factors transcribed by CREB and upregulates hippocampal synaptic plasticity [48–51], subsequently enhancing cognitive function [52, 53].

An additional mechanism of action of lithium for AD is the inhibition of cyclin-dependent kinases (cdk5) [54]. The enzymes are involved in cellular morphology, neuronal outgrowth, and tau phosphorylation in AD mouse models [55] and AD patients [54, 56–58]. The levels of these enzymes are elevated in AD [59–63]. Both cdk5 and GSK-3β are proline-directed serine/threonine kinases. It has been reported that GSK-3 and cdk5 are interconnected with dynamic crosstalk regulating each other's activities on aging and AD [54, 64, 65]. Interestingly, lithium has been shown to inhibit both kinases, and diminish tau phosphorylation, and accelerate apoptotic process and cell death in senescence-accelerated mice [66].

Another possible mechanism is to the enhancement of autophagy [67–69]. Autophagy is a key regulator in maintaining cellular homeostasis by degrading cellular components including p-tau and Aβ proteins [70, 71]. Since p-tau and Aβ are mainly degraded by the autophagy systems, autophagy dysfunction can lead to p-tau deposits [68]. It has long been known that autophagy dysfunction is an important mechanism underlying the neuropathology of AD [72, 73]. A recent study revealed that enhanced autophagy abolished tau phosphorylation and reversed memory impairment in transgenic AD mice [74]. Interestingly, lithium is a well-known autophagy enhancer in the mTOR-independent pathway by inhibiting GSK-3β [69]. Thus, the complex mechanisms of action related to GSK-3 inhibition are involved in an interrelated manner, explaining how lithium reduces tau, amyloid, and neuronal cell death and improve synaptic plasticity and cognitive function in AD.

Recent studies suggest the stabilization of aberrant Ca^{2+} signaling from the endoplasmic reticulum (ER) as the early mechanism of action of lithium for AD. Lithium is known to IMPase and IPPase, the key regulating enzymes in the phosphatidyl inositol cycle for recycling IP3. It can stabilize aberrant Ca^{2+} ER signaling by depleting supply of IP3-dependent receptor (IP_3R) to IP_3R-gated calcium channel at ER [11, 75]. We will discuss this issue further in the following sections.

Calcium Hypothesis of AD

Although the amyloid hypothesis has been the dominant theory of AD research over 3 decades, it has been challenged for its implications in late-onset AD and related dementia. Aβ plagues are often detectable in aging and in AD patients, and neurotoxic actions of Aβ peptides have been extensively demonstrated in vitro and in vivo investigations. Therefore, it was an unexpected surprise that anti-Aβ antibody or reagents can effectively remove Aβ depositions from the AD patient's brain but generally failed to show significant improvements of cognitive functions [18, 76–79]. Accumulating observations show that Aβ cascade pathology is poorly correlated with symptoms of AD such as progressive memory loss and cognitive decline in AD animal models and individuals with AD [80–82]. For example, aging individuals may have significant Aβ plaques and tau tangles without showing the clinically significant signs of cognitive deficits [75, 80, 81]. Recently, FDA has approved Aβ-directed monoclonal antibodies including aducanumab and lecanemab [83, 84]. It may need further studies to demonstrate whether these drugs effectively and safely improve cognitive function in AD patients [3, 4, 85].

Lately, more attention has been drawn to explore additional and alternative pathogenic mechanisms for late-onset AD [18, 77–79]. Among them, the hypothesis that Ca^{2+} dysregulation is the early pathophysiological mechanism of AD has gained enforced recognition recently [86–92]. This hypothesis has been reinforced by growing evidence that neuronal Ca^{2+} dysregulation is likely to be an initial pathogenic mechanism of AD [87, 93–96]. The Ca^{2+} hypothesis predicts that a moderate but sustained dysregulation of intracellular Ca^{2+} homeostasis could lead to AD pathological cascades, resulting in loss of synapses and neurons and subsequent decline of cognition [90, 91, 96]. In fact, perturbation in intracellular Ca^{2+} has long been demonstrated in imaging studies using cells and organoids from AD patients [97–99].

Early Mechanism of Action of Lithium on AD: Stabilizing Intracellular Ca^{2+} Signaling

A growing body of evidence suggests that lithium may stabilize disrupted intracellular Ca^{2+} signaling at ER as it's early mechanism of anti-AD effects. ER is a major intracellular Ca^{2+} calcium reservoir and plays a key role in maintaining intracellular Ca^{2+} homeostasis [100]. Inositol triphosphate-dependent receptor (IP_3R) and ryanodine receptors (RyR) are the Ca^{2+}-permeating channels regulating Ca^{2+} release from ER. The multiple lines of evidence suggest that the dysregulation of Ca^{2+} release from ER can result in formation of Aβ and p-tau [101, 102], disruption in synaptic plasticity, and cognitive deficits in AD [103–105]. Ca^{2+} imaging studies using AD animal models and mutant cell lines have shown that presenilin 1 (PS1) and presenilin 2 (PS2) mutations linked to familial Alzheimer's disease are associated with excessive ER Ca^{2+} response mediated by IP_3R [105, 106]. Thus, the IP_3R-mediated dyshomeostasis of intracellular Ca^{2+} may contribute to an early neuropathogenic mechanism of AD [90, 91, 107]. In two AD models including PS1 mutant mice and transgenic 3xTg-AD mice, the genetic reduction of IP_3R expression normalized exaggerated ER Ca^{2+} signaling, attenuated Aβ accumulation and tau hyperphosphorylation, and rescued impaired hippocampal LTP and memory deficits [108]. Furthermore, Wang et al. [109] reported that, in APP/PS1 AD mice, the subchronic treatment with xestospongin C (XeC), a reversible IP_3R antagonist, reduced Aβ plaques and ER stress proteins and improved cognitive behavior. Thus, it became increasingly clear that normalizing dysregulated ER calcium signaling via IP_3R is likely to be associated with therapeutic action against the major primary neuropathology of AD, leading to improvement of cognitive function.

The signal transduction pathway involving the phosphatidylinositol cycle is associated with IP_3R-mediated ER Ca^{2+} regulation and is affected by lithium. IP_3 is produced by the hydrolysis of a membrane phospholipid, phospholipase phosphoinositide 4,5-bisphosphate (PIP_2) after a primary messenger binds to the G-protein coupled receptor [110]. IP_3, as a second messenger, binds to IP_3R on the ER membrane and activates ER Ca^{2+} release. IP_3 is recycled through the phosphatidyl inositol cycle pathway. In the recycling cycle, IMPase is required for the de novo synthesis of inositol [111]. Lithium directly inhibits IMPase [112, 113] as well as IPPase [114, 115] and reduces IP_3 production [116, 117]. Thus, lithium depletes IP_3 supply to IP_3R at ER and downregulates ER Ca^{2+} release [11].

Table 2. The Effects of 30-day lithium treatment on aberrant Ca^{2+} signaling, nNOS expression, tau phosphorylation, and synaptic plasticity in 3xTg-AD mice [38]

	3xTg-AD mice		Non-Tg mice	
	lithium	control	lithium	control
Hippocampus (CA1)				
IP3-Ca^{+} response	↓	↑	→	→
VGCC-Ca^{+} response	↓	↑	→	→
Hippocampus (DG, CA1, CA2, CA3)				
nNOS	↓	↑	→	→
p-tau	↓	↑	→	→
Cortex				
nNOS	↓	↑	→	→
p-tau	↓	↑	→	→
Hippocampus (CA1-CA3)				
PTP	↑	→	↑	→
LTP	→	→	→	→

Our recent study has revealed that lithium stabilizes dysregulated Ca^{2+} release via ER acting at IP_3R. Specifically, subchronic lithium treatment normalizes aberrant IP_3R-mediated ER Ca^{2+} signaling in CA1 pyramidal cells in the hippocampus of 3xTg-AD mice. Interestingly, the same lithium treatment also reduces voltage-gated calcium channel (VGCC)-mediated Ca^{2+} influx in the CA1 hippocampal neurons in the AD mice [38] (Table 2). These effects of lithium treatment on dysregulated Ca^{2+} are largely associated with normalizing elevated neuronal nitric oxide (NO) synthase (nNOS) in the hippocampus and cortex in the 3xAD animal model. nNOS converts L-arginine to NO and L-citrulline [118, 119]. nNOS is largely activated by Ca^{2+} influx through N-methyl-D-aspartate (NMDAR) [120, 121] and VGCC [122, 123]. NO, a retrograde gaseous signaling molecule, is synthesized in postsynaptic dendrites and diffuses to presynaptic terminals, where it potentiates glutamate release and enhances Ca^{2+} influx mainly through NMDAR [124, 125]. VGCCs play an important role in maintaining normal synaptic transmission and intracellular signaling cascades by regulating Ca^{2+} influx [126]. Thus, the increased presynaptic glutamate release induces more Ca^{2+} influx via calcium channels such as NMDAR and VGCC [120, 121] and elevates NO levels by activating nNOS. Chronically sustained increases in NO, an oxidative-free radical, can exert detrimental effects such as oxidative-free radial neurotoxicity, Aβ formation, tau phosphorylation, mitochondrial stress, apoptosis, and synaptic dysfunction [44, 127–129]. The sustained release of excessive presynaptic glutamate with elevated NO may establish a pathogenic feedforward cycle of Ca^{2+} dysregulation contributing to the development of AD neuropathology leading to cognitive dysfunction [87, 130]. It has been shown that nNOS is increased in AD animal models and AD patients [108, 127, 131, 132]. Although the exact mechanism by which lithium normalizes elevated nNOS level is not fully understood, it has been shown that lithium inhibits the NMDA/NO pathway by inhibiting NMDAR activities [24, 133, 134]. nNOS can also be activated by Ca^{2+} influx through VGCC [122, 123]. Thus, it is likely that lithium suppresses nNOS expression by inhibiting Ca^{2+} influx via not only NMDAR but also VGCC. Thus, lithium may reduce NO-free radical-induced neurotoxicity by suppressing nNOS activity through inhibiting Ca^{2+} influx via NMDAR and VGCC. Our study also revealed that lithium treatment increased post-tetanic potentiation (PTP), a subtype of presynaptic short-term potentiation (Table 2). AD-mediated Ca^{2+} dysregulation can lead to deficits in Ca^{2+}-dependent plasticity including short-term plasticity such as PTP. PTP is required to synchronize network propagation and set LTP thresholds and enable short-term memory [135, 136].

Our studies suggest that the homeostatic action of lithium regulates aberrant Ca^{2+} signaling via multiple mechanisms. Lithium may break a chronic feedforward cycle of excessive Ca^{2+} influx by normalizing elevated NO production in AD. These actions together may lead to attenuating downstream neuropathogenic AD cascades such as oxidative-free radical neurotoxicity, tauopathy, neuronal loss, and impaired synaptic plasticity. Furthermore, lithium directly inhibits GSK-3β and cdk5, which significantly contribute to AD neuropathology. Thus, lithium may produce broad therapeutic actions against AD neuropathology at multiple levels from Ca^{2+} dysregulation through impaired synaptic plasticity, leading to the therapeutic actions against AD (Fig. 1). Our study has examined the subchronic (4 weeks) effects of lithium on Ca^{2+} dysregulation and AD neuropathology using a 3xTg-AD mouse model. To further pursue, the therapeutic role of lithium on stabilizing Ca^{2+} dysregulation as the early therapeutic mechanism of action against AD, the study needs to be extended to examine more chronic effects of lithium on Ca^{2+} dysregulation and subsequent neuropathologic process, impaired synaptic plasticity, and cognitive dysfunction. In addition, future studies should consider nonconventional AD models, such as mice with deficiency of NMDA receptor subunit GluN3A,

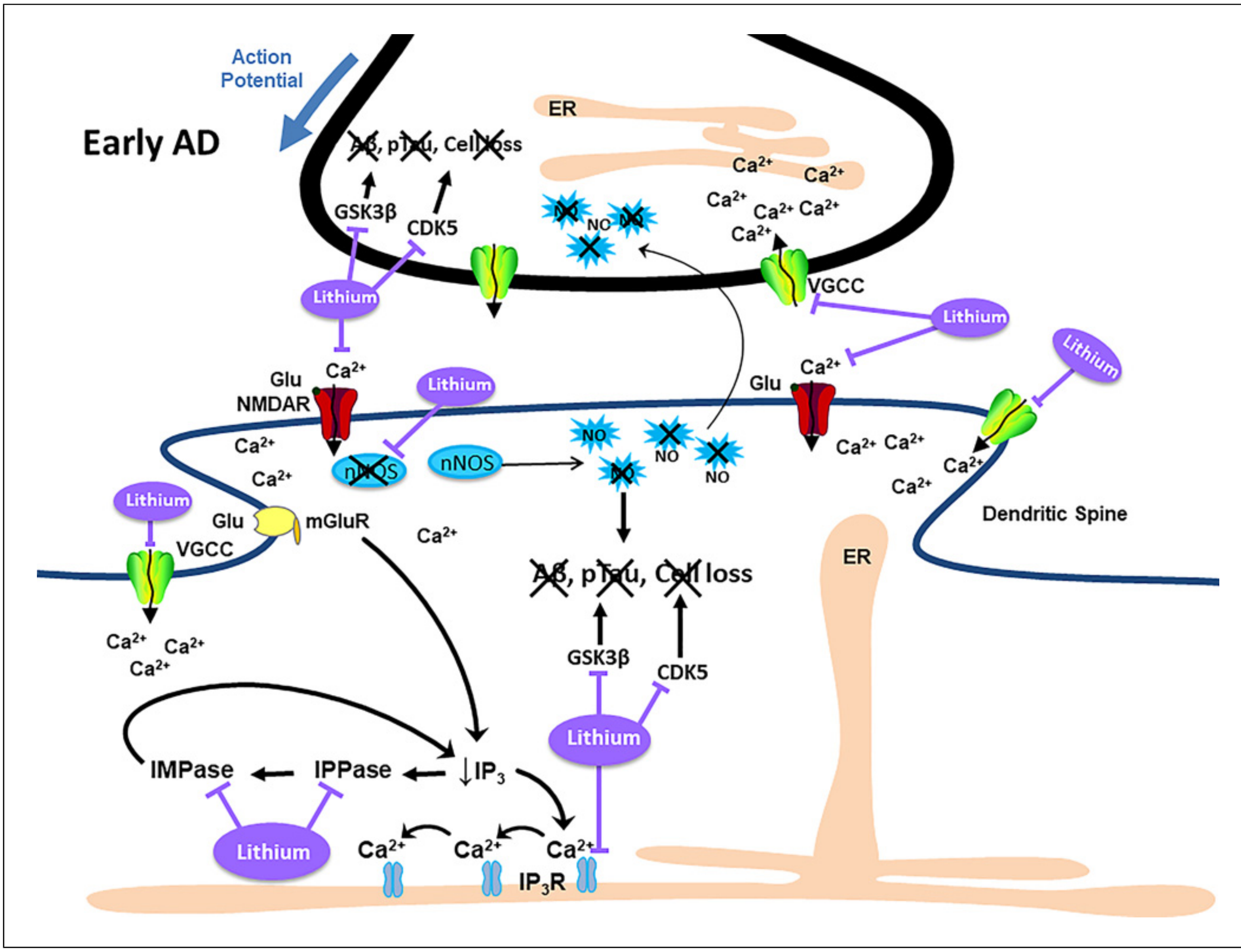

Fig. 1. Proposed mechanism of lithium's effects in AD. Lithium stabilizes elevated IP_3R-dependent ER Ca^{2+} release and VGCC-mediated Ca^{2+} influx in CA1 pyramidal neurons from early 3xTg-AD mice. Lithium also reduces nNOS expression and tau hyperphosphorylation in the hippocampus and cortex in the AD mice. The activation of nNOS is largely triggered by Ca^{2+} influx through NMDAR and VGCC. NO is synthesized by nNOS in postsynaptic dendrites and diffuses to presynaptic terminals, where it potentiates glutamate release and enhances postsynaptic Ca^{2+} influx. The increased glutamate release enhances Ca^{2+} influx via NMDAR and VGCC and elevates NO level, establishing the pathogenic feedforward cycle of AD leading to oxidative-free radical neurotoxicity. Lithium may block the pathogenic feedforward cycle by normalizing elevated nNOS expression. Lithium produces therapeutic actions against AD neuropathological cascades including p-tau, Aβ, neuronal cell death by stabilizing dysregulated Ca^{2+} signaling acting at nNOS expression, VGCC, and IP_3R-ER calcium signaling. GSK-3β and CDK5 are known to interact, leading to tau hyperphosphorylation, Aβ formation, and neuronal loss. Lithium may reduce tau phosphorylation by directly inhibiting GSK-3β and CDK5 in addition to stabilizing dysregulated Ca^{2+} signaling.

which causes calcium dysregulation and AD-like behavioral phenotypes [92]. Moreover, it would be interesting to study the intricate relationships of cytosolic and ER calcium during pathogenesis of AD through recently developed, red-shifted calcium indicator protein for ER [137]. We hope that such studies will establish and generalize lithium as an effective therapy for AD.

Clinical Studies on Therapeutic Effects of Lithium on AD

Clinical trials and epidemiological analyses suggest that lithium is likely to be efficacious in treating early-stage AD and MCI and reducing the risk of AD [7] (Table 3). A couple of retrospective medical record review studies suggest that patients with dementia or AD may benefit from lithium reported that patients with bipolar disorder or other mood disorders taking the standard doses of lithium for bipolar disorder (0.66–0.8 mEq/L) may have the less risk for dementia [138–140]. Two double-blind, placebo-controlled trials of lithium have shown that the long-term treatment of lithium over 12 months improved the cognitive symptoms in patients with MCI and slowed down the progression of MCI to AD [141, 142]. Note that lithium levels in the blood in these studies (0.25–0.5 mEq/L) were lower than the therapeutic window of blood levels for bipolar disorder (0.5–0.9 mEq/L). Interestingly, a double-blind, placebo-controlled trial of trace dose of lithium (300 μg/day as compared to the range of lithium doses for bipolar disorder of 600–1,200 mg/day) for 15 months reported that lithium improved cognitive symptoms in patients with dementia [143]. At the trace dose, the blood level of lithium is not detectable. Lithium is known to have multiple adverse effects in the chronic treatment [144]. Since the risk of the adverse effects of lithium is significantly reduced at the low blood levels of lithium [145], these studies suggest that the lower or trace doses of lithium would improve cognitive symptoms without the substantial risk of the adverse effects of lithium. A single-blind, placebo-controlled study reported that lithium treatment less than 8 months did not improve cognitive symptoms in AD patients [146], suggesting that the duration of the lithium treatment may not be long enough to produce significant therapeutic. These trials together suggest that the dose of lithium could be lower than the therapeutic window for bipolar disorder, but the duration of trial would be long enough to produce the therapeutic effects of lithium on dementia.

Epidemiological studies generally support the clinical benefits of lithium for dementia [7]. The epidemiological studies of retrospective registration reviewing a very number of general population have reported that lithium is likely to reduce the risk for AD [147–149, 152]. Kessing et al. [147] conducted an epidemiological study with population-based analysis of over 40,000 individuals to compare individuals taking lithium and those not taking the drug as a 10-year follow-up study. The authors reported that individuals taking lithium at least more than once were associated with the decreased rate of dementia. The authors also analyzed nearly 5,000 individuals to compare individuals with bipolar disorder taking lithium with those taking anticonvulsants, antidepressants, and/or antipsychotics [148]. They concluded that long-standing lithium treatment was associated with a less risk for dementia by comparing those with taking other medications. Gerhard et al. [149] conducted a cohort study with the analysis of over 40,000 individuals older

Table 3. Clinical trials and epidemiological studies of lithium for Alzheimer's disease

Study	Design	Subject numbers	Length of study	Li. level mEq/L	Type of dementia	Results
Studies with clinical lithium levels (therapeutic window: 0.6–1.2 for bipolar disorder)						
Forlenza et al. [141] 2011	Double-blind, placebo	45	12 months	0.25–0.5	aMCI	↑*
Forlenza et al. [142] 2019	Double-blind, placebo	61	24 months	0.25–0.5	aMCI	↑
Terao et al. [140] 2006	Double-blind, placebo	60	6 months	0.66, average	Early AD	→
Hampal et al. [146] 2009	Single-blind, placebo	71	6 months	0.5–0.8	Early AD	→
Studies on trace lithium dose						
Nunes et al. [143] 2013	Double-blind, placebo	113	15 months	300 μg/day	Dementia	↑
Epidemiological studies						
Kessing et al. [147] 2008	Individuals W/O Li	16,238	10 years		Dementia	↑
Kessing et al. [148] 2010	Individuals with bipolar W/O Li	4,856	10 years		Dementia	↑
Gerhard et al. [149] 2015	Individuals with bipolar W/O Li	41,931	301–365 days		Dementia	↑
Angst et al. [138] 2007	Individuals with mood disorders W/O Li	406	4 years	0.7, average	Dementia	↑*
Kessing et al. [150] 2017	Population drinking water	1,193		11.5 μg/L, mean	AD	↑
Fajardo et al. [151] 2018	Population drinking water	6,180		56u/L, mean	AD	↑

↑, Improve cognitive function or lower risk of dementia. *Inverse correlation between lithium and severity of dementia.

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than 50 with bipolar disorder taking lithium and those without lithium. The authors reported that individuals taking lithium only for 301–365 days, but not those less than the days, were associated with a significantly reduced risk for dementia, emphasizing the importance of the treatment of lithium for a prolonged period of time in elderly individuals to effectively reduce the risk of dementia. However, a population-based nested case-control study analyzing population older than 65 years old including over 40,000 individuals with AD reported that lithium treatment did not increase or decrease the risk for AD [152]. The authors suggested further investigation to consider the influence of multiple confounding factors.

Two epidemiological studies compared general populations who were exposed to the micro or trace doses of lithium in drinking water and those not exposed to lithium in multiple regions [150, 151]. Kessing et al. [150] reported that the incidence rate of dementia was generally decreased in those exposed to lithium. These findings were consistent with a study, which reported that the mortality rate of AD was negatively correlated with trace lithium levels in drinking water [151]. Although conclusive evidence for the efficacy of lithium is limited, the clinical trials along with the epidemiological analyses suggest that lithium is likely to be efficacious and safe for individuals with AD and AD-related dementia.

Conclusion

Preclinical studies have shown that lithium reduces amyloid and tau pathology, attenuates neuronal loss, enhances synaptic plasticity, and improves cognitive function in various AD animal models. The inhibition of GSK-3β and cdk5 is likely to be associated with these therapeutic actions. Most of these studies have shown that lithium acts on multiple therapeutic targets, but they have not thoroughly investigated upstream mechanisms driving these therapeutic actions. Our recent study revealed that the stabilization of dysregulated Ca^{2+} homeostasis may attenuate downstream neuropathogenic AD processes such as oxidative-free radical neurotoxicity, amyloid and tau pathology, and impaired synaptic plasticity. Through these therapeutic actions, lithium may produce broad therapeutic actions against AD with potential to modify the disease process. Consistent with the preclinical studies, clinical studies have shown that lithium reduces the risk of AD and deters the progress of MCI and early AD. Thus, the preclinical and clinical studies together strongly suggest that lithium is likely to be efficacious as a disease-modifying drug by acting on the multiple therapeutic targets of AD.

Conflict of Interest Statement

Authors have no conflict of interest to declare.

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Author Contributions

The authors confirm significant contribution to the article. All the authors participated in the study conception and design and data collection and analysis substantially. Seong Shim drafted the manuscript preparation. Ken Berglund and Shan Ping Yu provided critical reviews and collaborated in revising the manuscript multiple times. All the authors approved the final version of the manuscript.

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